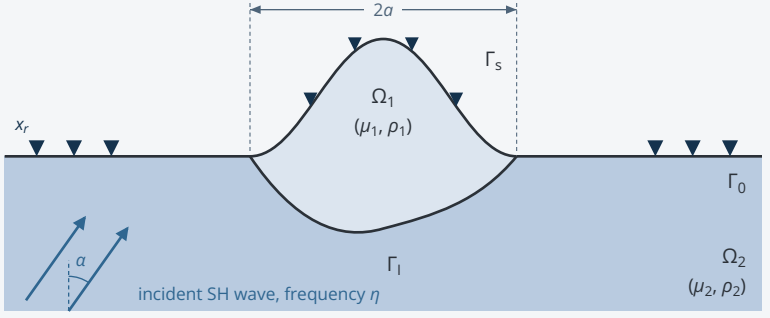
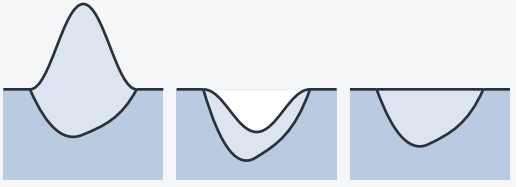
System and notation



Types of irregularity

Ω_1 lies between Γ_s and Γ_I; $\Gamma_s \cup \Gamma_0$ is traction-free

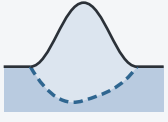


Hill
Γ_s above Γ_0

Canyon
Γ_s below Γ_0

Sedimentary basin
flat Γ_s on Γ_0

Inverse problems



Interface identification

Unknown: Γ_I

Receivers on $\Gamma_0 \cup \Gamma_s$

Surface reconstruction

Unknown: Γ_s

Receivers on Γ_0

Joint identification

Unknown: Γ_s, μ_1, ρ_1

Receivers on Γ_0

known · unknown · receivers · incident wave · irregularity Ω_1 · unknown material · half-space Ω_2